

Camera Slider

Wiring & Connection Reference

Firmware v1.0

Component	Part	Interface
Microcontroller	Arduino Uno	—
Stepper Driver	TMC2209	STEP / DIR / EN
Stepper Motor	NEMA 17	4-wire to TMC2209
Display	SH1106 1.3" OLED	I2C (A4/A5)
Encoder	EC11 + 2 buttons	Digital (D2–D6)
IR Receiver	Standard module	Digital (D7)
Endstops	2 x NC switch	Digital (D12/D13)
Buttons	Left / Right / E-Stop	Analog (A0–A2)

All buttons wired to INPUT_PULLUP — connect between pin and GND. Endstops are Normally Closed (NC).

1 Power Supply

Rail	Source	Connects To	Notes
5 V (logic)	USB or Uno 5V pin	Arduino Uno	Powers logic circuits
3.3 V	Uno 3.3V pin	OLED VCC	Max 50 mA draw
12–24 V (VM)	External PSU	TMC2209 VM	Motor supply — match motor spec
GND	Common	All components	Star topology recommended

Keep motor power (VM) wiring away from signal lines to reduce noise.

2 OLED Display — SH1106 1.3" (I2C)

OLED Pin	Arduino Pin	Notes
VCC	3.3 V	Do NOT connect to 5 V — display is 3.3 V only
GND	GND	
SDA	A4	I2C data — shared bus
SCL	A5	I2C clock — shared bus

Default I2C address: 0x3C. If display does not initialise, try 0x3D.

3 Rotary Encoder — EC11 (on OLED board)

EC11 Pin	Arduino Pin	Notes
CLK (A)	D2	Hardware interrupt pin — required for reliable decoding
DT (B)	D3	Encoder direction channel
SW (button)	D4	Encoder push — active LOW (INPUT_PULLUP)
VCC	5 V	
GND	GND	

4 Encoder Board Buttons (on OLED board)

Button	Arduino Pin	Function	Notes
BACK	D5	Back / cancel	Active LOW — INPUT_PULLUP
CONFIRM	D6	Select / confirm	Active LOW — INPUT_PULLUP

5 IR Receiver Module

IR Module Pin	Arduino Pin	Notes
DATA (OUT)	D7	Signal output

VCC	5 V	
GND	GND	

Use `IR_Learn.ino` to discover button codes, then paste into the `IR_BTN_*` defines in `camera-slider.ino`.

6 Stenner Driver — TMC2209

TMC2209 Pin	Arduino Pin / Supply	Notes
STEP	D9	Step pulse — one pulse per microstep
DIR	D10	Direction — HIGH or LOW
EN	D11	Enable — LOW = active, HIGH = disabled
VIO	5 V	Logic supply
VM	12–24 V PSU	Motor power supply
GND	GND (both grounds)	Logic GND and power GND joined at PSU
PDN_UART	Not connected	Leave floating for standalone mode

Set TMC2209 microstepping via `MS1/MS2` pads on the module. Default in firmware: 16 microsteps (`STEPS_PER_MM = 80`). Adjust `STEPS_PER_MM` in `camera-slider.ino` to match your lead-screw pitch.

7 Stenner Motor — NEMA 17

Motor Wire	TMC2209 Terminal	Notes
Coil A+	A1 (or M1)	Coil A — check motor datasheet for colour
Coil A-	A2 (or M2)	
Coil B+	B1 (or M3)	Coil B
Coil B-	B2 (or M4)	

If the motor runs in the wrong direction, swap either the A pair (A+ / A-) or the B pair (B+ / B-) — not both.

8 Endstops (Normally Closed)

Endstop	Arduino Pin	Wiring	Notes
Left (home)	D12	One wire → D12, other → GND	Triggers LOW when carriage reaches home
Right (limit)	D13	One wire → D13, other → GND	Triggers LOW at far end of travel

NC (Normally Closed) wiring is safer — a broken wire triggers the stop. Set `ENDSTOP_TRIGGERED` to HIGH in firmware if using NO (Normally Open) switches.

9 External Buttons

Button	Arduino Pin	Function	Notes
LEFT jog	A0	Jog carriage left	Tap = 1 mm step Hold = fast jog

RIGHT jog	A1	Jog carriage right	Tap = 1 mm step Hold = fast jog
E-STOP	A2	Emergency stop	Controlled decel then motor disable

All buttons: connect between pin and GND. Arduino INPUT_PULLUP provides the pull-up resistor — no external resistor needed.

10 IR Remote — Button Function Map

Define in firmware	Suggested remote key	Function
IR_BTN_START_PAUSE	OK / Play	Start, pause, or resume active program
IR_BTN_CANCEL	Stop	Cancel paused program, return to start
IR_BTN_HOME	Home / 0	Run homing sequence
IR_BTN_FULL_LOOP	* or #	Select Full Loop program
IR_BTN_SLOT_1	1	Select user program slot 1
IR_BTN_SLOT_2	2	Select user program slot 2
IR_BTN_SLOT_3	3	Select user program slot 3
IR_BTN_SLOT_4	4	Select user program slot 4
IR_BTN_LEFT	Left arrow	Jog carriage left (single slow step)
IR_BTN_RIGHT	Right arrow	Jog carriage right (single slow step)

Upload `IR_Learn.ino`, open Serial Monitor at 115200 baud, and press each button to read its hex code. Paste codes into the `IR_BTN_*` defines near the top of `camera-slider.ino`.

11 I2C Bus Summary

Device	I2C Address	SDA Pin	SCL Pin
SH1106 OLED	0x3C	A4	A5

Only one I2C device on this bus. If adding more devices ensure each has a unique address. Use 4.7 kOhm pull-up resistors on SDA and SCL if bus length exceeds ~20 cm.

12 Arduino Uno — Complete Pin Summary

Pin	Direction	Connected To	Function
D2	Input	EC11 CLK	Encoder channel A (interrupt)
D3	Input	EC11 DT	Encoder channel B
D4	Input	EC11 SW	Encoder push button
D5	Input	BACK button	Menu back / cancel
D6	Input	CONFIRM button	Menu select / confirm
D7	Input	IR receiver DATA	Infrared remote input

D8	—	Not connected	Reserved
D9	Output	TMC2209 STEP	Stepper step pulse
D10	Output	TMC2209 DIR	Stepper direction
D11	Output	TMC2209 EN	Stepper enable (LOW = on)
D12	Input	Left endstop	Home / left limit switch
D13	Input	Right endstop	Right limit switch
A0	Input	LEFT jog button	Carriage jog left
A1	Input	RIGHT jog button	Carriage jog right
A2	Input	E-STOP button	Emergency stop
A3	—	Not connected	Reserved for future use
A4	I2C SDA	OLED SDA	I2C data
A5	I2C SCL	OLED SCL	I2C clock

D0 and D1 are used by Serial (USB) — do not connect hardware to these pins.